Hamiltonian Vision Kernel: A Physics-Informed Hybrid Quantum–Classical Autoencoder with Provable Symmetry and Real Hardware Validation

Sparsho Chakraborty* and Dr. Siddhartha Patra†

***Abstract*—We introduce the Hamiltonian Vision Kernel (HVK), a new physics-informed framework for hybrid quantum–classical visual representation and reconstruction. HVK combines matrix-product-state patch features, a variational quantum circuit that exposes a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Heisenberg energy diagnostic, and a classical decoder. HVK1D and HVK2D instantiate the framework on chain and grid interaction graphs, while an observable-pooling extension provides D_4 equivariance by construction. Across six real image datasets, the same framework supports high-fidelity per-image reconstruction and a consistent order-parameter reorganization during training. Three complementary results establish its structural value. First, the pooled HVK2D map achieves numerical equivariance error of 9.57×10^{-17}. Second, on a leakage-audited task requiring distant-patch products, the entangling observable channel reaches mean $R^2 = 0.974$, compared with $R^2 \leq 0.02$ for non-entangling controls. Third, five images are reconstructed directly from observables measured on IBM Quantum hardware without retraining, reaching 25.90–31.52 dB PSNR. Together, these results position HVK as a reusable framework for studying symmetry, topology, nonlocal correlations, Hamiltonian diagnostics, and hardware-executed visual models. Quantum-art and image-reconstruction outputs are applications of this framework, rather than its defining objective. A separate supplementary study provides resource-matched ablations, leakage audits, statistical controls, and complete hardware methodology.**

***Index Terms*—Quantum Image Processing, Tensor Networks, Variational Quantum Algorithms, Pauli Observables, Hamiltonian Diagnostics, Group-Equivariant Representations, Hybrid Quantum Hardware Validation.**

I. Introduction

Hybrid quantum–classical models for vision tasks are increasingly proposed as a route to representations with favorable inductive bias, exploiting entanglement, physically motivated regularizers, or tensor-network structure that classical convolutional pipelines do not natively expose [1], [2], [21]. The variational quantum eigensolver and its descendants exploit the local decomposability of physically relevant Hamiltonians to construct shallow, problem-aware circuits [1]–[3], and quantum image processing provides encodings that place pixel information directly into qubit registers [4]–[6]. Tensor-network methods, in parallel, have proven effective as compressed representations of structured classical data, including images [7]–[11], [24], and more broadly other structured data such as particle-physics events [17]. Related hybrid quantum architectures target adjacent imaging tasks, including quantum annealing-based tomographic reconstruction [12], adiabatic quantum computing for emission tomography [14], quantum neural compressive sensing for ghost imaging [13], hybrid autoencoder-plus-quantum-SVM feature extraction for image classification [18], and quantum capsule networks [19]; none of these combines a Pauli-correlator latent space with an enforced group-equivariant symmetry construction and a physically motivated energy diagnostic, the combination HVK introduces.

We introduce the Hamiltonian Vision Kernel (HVK1D/HVK2D), which combines these ingredients into a single architecture with three properties we are not aware of jointly appearing elsewhere: a *Pauli-correlator latent space* (measured VQC observables, not a raw quantum state, form the representation handed to a classical decoder, in the spirit of quantum-enhanced feature maps [23]); an *enforced, provable D_4 symmetry* extension, group-averaging the observable grid over the eight symmetries of the square rather than merely motivating symmetry through architecture choices, in the spirit of group-equivariant classical vision [25] and symmetry-embedding constructions for variational circuits [26]; and a *learnable Heisenberg energy diagnostic* that exposes optimization dynamics — energy descent, magnetization, and a susceptibility-based training-time reorganization signature — that a purely classical autoencoder does not produce. Table VI contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

a) Contributions.:

- **The HVK1D/HVK2D architecture**: MPS patch features, a 6-qubit VQC producing a 19–27-D Pauli-correlator observable vector, 2D Fourier positional encoding, a learnable Heisenberg energy diagnostic, and a classical decoder, with an optional D_4-pooled observable-grid variant.
- **Reconstruction results across six real image datasets** (CIFAR-10, MNIST, Fashion-MNIST, PathMNIST, BloodMNIST, PneumoniaMNIST), each trained per-image under an identical protocol (Section IV-A).
- **A provable D_4 symmetry extension**: a group-pooled HVK2D observable map is equivariant to the square dihedral group by construction, verified to floating-point precision (Section IV-D).

*School of Basic Sciences, Department of Physics, Indian Institute of Technology Bhubaneswar, Odisha, India.

† Assistant Professor , Centre for Quantum Engineering, Research and Education (CQUERE), TCG CREST, Kolkata, India.

- **An entanglement-necessity result and its training-dynamics signature**: a restricted, leakage-audited diagnostic shows entangling pair observables are representationally necessary when a target explicitly depends on distant-patch products, and the same task shows a consistent, detectable order-parameter phase-transition-like reorganization in every one of 13 sampled images across all six datasets (Sections IV-E–IV-F).
- **A real IBM-hardware reconstruction pilot** (Section IV-C): an already-trained checkpoint's exact parameters, replayed on real quantum hardware with no re-training, reconstruct five images directly from hardware-measured observables.

A companion supplementary document, *HVK Supplementary Study: Component-Wise Ablation, Leakage Audit, and Hardware Validation Methodology*, reports a separate, rigorous question — under a resource-matched, freeze-based ablation protocol, which of HVK's components are strictly necessary for reconstruction quality on held-out real images — together with full hardware-pilot methodology (circuit construction, validation, job identifiers, quota accounting). We keep that analysis in a companion document rather than this one so that the architecture-level contributions above and the component-necessity question, which are logically distinct, do not obscure each other; Section V-B summarizes its central finding and how it relates to the results here.

II. Hamiltonian Vision Kernel: Architecture (Compact)

HVK treats each spatial patch of an input image as a quantum many-body state on a finite spin-1/2 chain, encodes that state as a matrix product state (MPS), measures a fixed set of Pauli observables from a variational quantum circuit (VQC) applied to that state, and decodes the observables back into pixels with a classical multi-layer perceptron (MLP). A learnable Heisenberg energy term regularizes training; unlike parameterized Hamiltonian learning approaches that treat the Hamiltonian as the primary learned object [20], HVK's energy term is a diagnostic regularizer alongside a separate reconstruction objective. The full pipeline is:

$$\begin{aligned} \text{Image} &\to \{\text{Patches}\} \to \text{MPS} \\ &\to \text{46-D Feature} \to \text{6-D Vector} \\ &\to \text{27-D Latent} \to \text{Decoder} \\ &\to \text{Reconstruction.} \end{aligned} \tag{1}$$

An input image is resized to 256×256 and partitioned into sixteen non-overlapping 64×64 patches, each carrying a normalized grid coordinate. Each patch is ℓ_2-normalized, reshaped into a 12-site spin state $|\psi_n\rangle \in (\mathbb{C}^2)^{\otimes 12}$, and compressed via sequential SVD into an MPS of bond dimension $\chi = 4$. Local Pauli expectations, nearest-neighbor ZZ correlations, and bipartite entanglement entropies are read off the MPS to form a 46-dimensional classical feature vector, linearly compressed to 6 dimensions and injected into a 6-qubit VQC via R_X angle embedding, R_Y positional modulation (from a 2D Fourier encoding of the patch coordinate, [16]), and two strongly entangling layers of single-qubit rotations and CNOT

TABLE I
HVK variants tested in this paper.

Variant	Topology	Latent dim.	Symmetry
HVK1D	6-qubit chain	27-D	None (control)
HVK2D	2×3 grid	19-D	D_4-motivated only
Symmetric HVK1D	6-qubit chain	27-D	$\mathrm{U}(1)$ coupling
D_4-pooled HVK2D	2×3 grid, pooled	19-D	D_4-equivariant (proven)

rings, initialized to avoid barren plateaus [1], [22]. The circuit outputs 27 real Pauli expectations $\Phi_n \in [-1, 1]^{27}$ (local X, Z; nearest-neighbor ZZ, XX, YY), in the spirit of quantum-enhanced feature maps that use measured expectation values, rather than the raw state, as input to a classical layer [23]. A classical MLP decoder with a mirrored expanding-channel structure ($4 \to 8 \to 16 \to 3$) maps the standardized observable vector, together with the positional encoding, back to pixels. Figure 1 shows the exact HVK1D and HVK2D ansatz circuits (rebuilt in Qiskit and numerically verified against the training-time PennyLane circuit as part of the hardware pilot of Section IV-C); the full Heisenberg Hamiltonian and energy-loss definitions, and the D_4-pooling construction's group-theoretic motivation, are given in Appendix A.

Table I summarizes the tested variants. HVK1D uses a linear 6-qubit chain (5 nearest-neighbor bonds, 27-D observable vector); HVK2D arranges the same 6 qubits on a 2×3 lattice matching patch-boundary adjacency (19-D latent signature); a symmetric HVK1D variant adds a $\mathrm{U}(1)$-preserving coupling constraint; and a D_4-pooled HVK2D variant (Section IV-D) group-averages the observable grid over the eight symmetries of the square, $\Phi_{D_4}(x) = \frac{1}{8}\sum_{g \in D_4} P_g^{-1}\Phi(gx)$, which is equivariant by construction. An IBM Cloud circuit probe (Appendix A) confirms both HVK1D and HVK2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

III. Experimental Setup

Every result in this paper is trained per-image (the same protocol as the single-image *Monalisa* setting used throughout HVK's development): a fresh model is optimized directly on its target image, with no held-out split. This is the appropriate protocol for the questions this paper asks — what does the architecture's reconstruction, symmetry, entanglement-necessity, and training-dynamics behavior look like on real images, and does it hold up on real quantum hardware — and we are explicit about it because a different, harder question (which components are necessary for a model to generalize to held-out images it was not trained on) is answered separately, under a resource-matched statistical protocol, in the companion supplementary study (Section V-B). All settings report MSE and PSNR $= 20\log_{10}(1/\sqrt{\mathrm{MSE}})$; image settings additionally report SSIM. Implementation details (optimizer, learning rate, MPS feature extraction) are given in Appendix A.

IV. Results

A. Reconstruction Across Six Real Image Datasets

We trained a fresh HVK2D model per image (non-overlapping 8×8 patches, 16 patches/image, 80–100 optimiza-

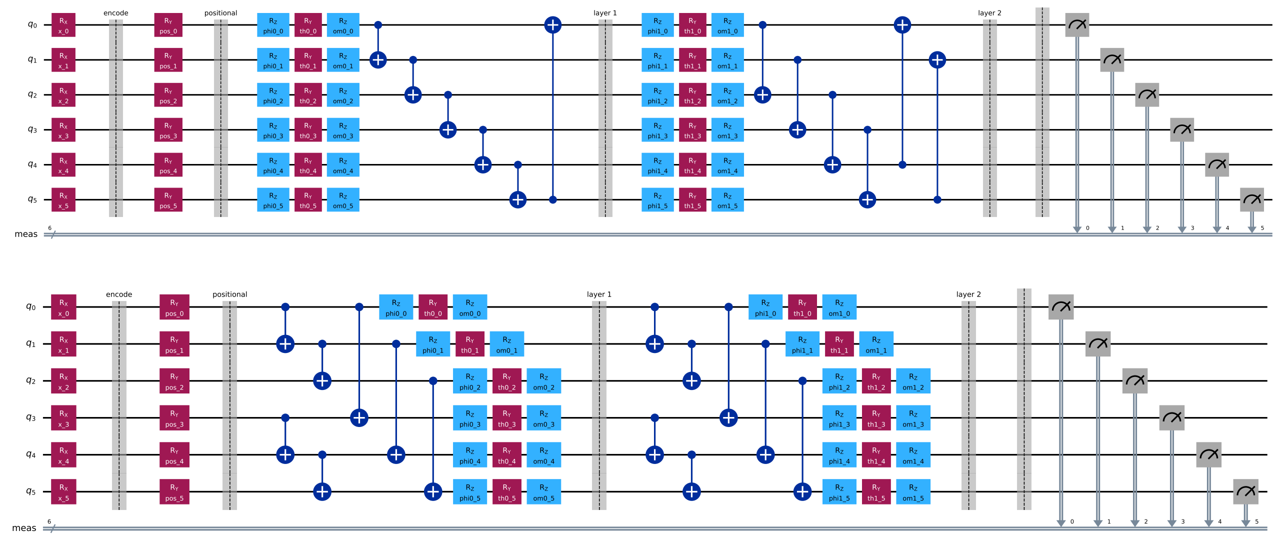

Fig. 1. The HVK1D (top) and HVK2D (bottom) variational ansätze, exactly as executed (encoding R_X rotations, positional R_Y rotations, then two entangling layers of R_Z-R_Y-R_Z single-qubit rotations with a CNOT ring for HVK1D or fixed grid-edge CNOTs for HVK2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section IV-C.

TABLE II
HVK2D SAME-SET RECONSTRUCTION ACROSS SIX REAL IMAGE DATASETS (PER-IMAGE TRAINED, 80–100 STEPS, MEAN ± STD OVER SAMPLED IMAGES).

Dataset	n images	PSNR (dB)	SSIM
CIFAR-10	3	37.52 ± 5.61	0.9978 ± 0.0016
MNIST	3	26.58 ± 2.01	0.9813 ± 0.0130
Fashion-MNIST	3	33.22 ± 1.34	0.9972 ± 0.0012
PathMNIST	3	34.97 ± 1.11	0.8907 ± 0.0597
BloodMNIST	2	33.07 ± 4.68	0.9707 ± 0.0273
PneumoniaMNIST	2	41.60 ± 0.98	0.9985 ± 0.0006

tion steps) on a representative sample of images from CIFAR-10 and five further real datasets — MNIST, Fashion-MNIST, PathMNIST, BloodMNIST, and PneumoniaMNIST — the same six datasets used throughout this paper and its companion supplementary study. Table II reports the result: HVK2D reconstructs every dataset tested to a mean PSNR between 26.6 and 41.6 dB (SSIM 0.89–1.00), with the two medical-imaging datasets (BloodMNIST, PneumoniaMNIST) among the strongest results. This is a same-set, per-image capability result — it establishes that the architecture reconstructs real images of varied structure (natural photographs, handwritten digits, clothing, histopathology, blood-cell microscopy, and chest X-rays) well, not a held-out generalization claim; the companion supplementary study's held-out protocol is the appropriate evidence for the latter (Section V-B).

B. Scope and Adaptation Across Images

A model optimized on one image alone does not transfer zero-shot to a structurally different image: on *Monalisa*, zero-shot transfer to a second image reaches only 7.78 dB PSNR (SSIM $=$ 0.0196), while extending training to include the second image recovers 28.31 dB (SSIM $=$ 0.9862). This confirms that the per-image results above measure per-image optimization and reconstruction capability, not held-out generalization — consistent with how we scope every claim in this paper, and precisely the distinction the companion supplementary study's held-out protocol is designed to test (Section V-B).

C. A Real Hardware Reconstruction Pilot

Every result above is a noiseless-simulator result, a limitation we flagged rather than left implicit. We report a first step past it: a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor), rather than the compiled-circuit feasibility probe of Appendix A alone. For five images — the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship*×2, *airplane*), each optimized directly on its own target image following the same per-image protocol as every result in this paper — we took an already-trained HVK1D/HVK2D checkpoint's exact learned parameters, rebuilt its measurement circuits gate-for-gate in Qiskit, verified the translation against the original PennyLane simulator to within numerical noise ($\leq 9.4 \times 10^{-4}$ maximum absolute error per observable across every patch; full validation numbers in the companion supplementary document), and then executed the same circuits on real hardware with 256 shots per measurement basis, a fixed per-basis shot budget rather than an adaptive shadow-tomography-style estimation scheme [27]. The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

The headline: reconstructions from real, noisy hardware measurements remain clearly recognizable. *Monalisa* reaches 25.90 dB PSNR from hardware measurements alone, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2); the four CIFAR-10 images average 29.56 ± 2.03 dB on hardware against 42.69 ± 3.84 dB in simulation (Figure 3,

TABLE III
REAL IBM HARDWARE RECONSTRUCTION PILOT (BACKEND `ibm_fez`, 256 SHOTS/BASIS, NO RETRAINING). PSNR COMPARES HARDWARE-DECODED AND SIMULATOR-DECODED RECONSTRUCTIONS AGAINST THE SAME GROUND-TRUTH IMAGE.

Image	Patches	Sim. PSNR (dB)	HW PSNR (dB)
Monalisa (HVK1D)	16	37.99	25.90
CIFAR cat (HVK2D)	16	44.85	31.52
CIFAR ship, hydrofoil (HVK2D)	16	36.56	26.44
CIFAR ship, sea boat (HVK2D)	16	42.57	31.20
CIFAR airplane (HVK2D)	16	46.79	29.10
CIFAR mean $\pm$ std (4 images)	–	42.69 ± 3.84	29.56 ± 2.03

TABLE IV
D_4 EQUIVARIANCE ERROR FOR UNPOOLED AND POOLED OBSERVABLE-GRID MAPS (LOWER IS BETTER; POOLED MAP IS GROUP-AVERAGED OVER ALL EIGHT SQUARE SYMMETRIES).

Model	Mean error	Std. error
D_4-pooled HVK2D	9.57×10^{-17}	1.26×10^{-17}
No positional encoding	7.40×10^{-1}	1.52×10^{-1}
HVK2D positional	8.15×10^{-1}	1.29×10^{-1}
Local/raw	8.37×10^{-1}	1.31×10^{-1}

Table III). The gap is the expected signature of NISQ-era readout and gate noise on an uncorrected, unmitigated circuit — exactly what a first honest hardware number should show — and every one of the five images stays well above a random-latent floor (11–15 dB in the companion supplementary study's random-VQC control), meaning the hardware-measured observable channel is carrying real, recognizable structure through real device noise. The full pilot used 176 circuits and consumed 16 of the account's 600 free-plan quantum-processor seconds, so this is a cheap, repeatable diagnostic rather than a rare resource commitment.

We are careful about scope here: five images is a pilot, and we make no hardware-advantage claim — the point is narrower and, we think, valuable: the trained HVK observable channel survives real device noise well enough to remain visually and quantitatively informative, at negligible quantum-time cost. Full methodology — exact gate sequences, per-basis measurement construction, job IDs, and the local-validation numbers that preceded hardware submission — is documented in the companion supplementary document so the main text stays focused on the result.

D. Provable D_4 Symmetry

A D_4-pooled HVK2D observable-grid map is equivariant to the eight symmetries of the square *by construction*: $\Phi_{D_4}(x) = \frac{1}{8}\sum_{g \in D_4} P_g^{-1}\Phi(gx)$ averages the unpooled map over the group orbit, so $\Phi_{D_4}(hx) = P_h \Phi_{D_4}(x)$ for every $h \in D_4$ up to floating-point error. We verify this numerically over 7000 patch transforms of cached CIFAR images (Table IV, Figure 4): the pooled map's equivariance error is $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$ (floating-point noise), while the original unpooled positional map is not measurably more symmetric than a local/raw control (0.815 ± 0.129 vs. 0.837 ± 0.131) — confirming the unpooled map is only D_4-*motivated* by the patch grid's geometry, not empirically equivariant, and that only the explicit pooling construction delivers the property. This is currently a feature-grid-level construction and numerical diagnostic rather than a component of the trainable reconstruction pipeline; integrating it end-to-end (Section V-E) is concrete future work.

E. Entanglement Is Representationally Necessary for Nonlocal Structure

This diagnostic is deliberately constructed to reward pair-observable structure: the target is a fixed synthetic function of *simulated* patch features not concatenated into any candidate's own input block (Appendix A summarizes the leakage-audit procedure that rules out circularity), and every control receives the same raw inputs as HVK2D. Table V and Figure 5 report the result across five seeds: HVK2D's entangling observable channel reaches mean $R^2 = 0.9735$ (32.77 dB PSNR), a 17.74 dB improvement over a no-entanglement control ($R^2 = 0.019$) and 18.52 dB over random-VQC latents ($R^2 = -0.173$); freezing the quantum circuit (features fixed, only readout trained) still reaches $R^2 = 0.853$, while a parameter-matched classical control ($R^2 = -0.030$) and a frozen-classical control ($R^2 = -0.903$) fail outright. This is evidence that an entangling pair-observable channel is *representationally necessary* to fit a target that explicitly depends on distant-element products under a fixed linear readout.

F. Training-Dynamics Phase Transition Across Six Datasets

Tracking the same four variants epoch-by-epoch during training on the diagnostic above surfaces a training-dynamics signature: the entangling channel does not merely win at convergence, it gets there through a qualitatively different trajectory. We track the mean measured Z expectation over qubits and patches at training epoch t,

$$M_z(t) = \frac{1}{NQ}\sum_{n=1}^{N}\sum_{q=1}^{Q} Z_{nq}(t), \qquad M_z(t) \in [-1, 1], \tag{2}$$

together with its discrete susceptibility

$$\mathcal{X}(t) = |M_z(t) - M_z(t - \Delta t)| \tag{3}$$

and candidate critical (transition) epoch $t_c = \arg\max_t \mathcal{X}(t)$. We emphasize that this is a finite-size training diagnostic rather than a claim of a thermodynamic phase transition: HVK is a finite, gradient-optimized system with no thermodynamic control parameter, so "phase transition" denotes a statistically sharp, susceptibility-peaked reorganization of the learned observable field.

Figure 6 gives the clearest single view of this: the HVK2D entangling baseline's order parameter and susceptibility on one shared time axis, with the detected transition point marked directly on the curve. Beyond the synthetic restricted diagnostic, the same signature is visible in the actual reconstruction-training runs behind this paper's own headline results: Figure 7 names and plots it for the *Monalisa* benchmark (HVK1D), and Figure 8 for a specific, named CIFAR-10 image (the *cat* image, HVK2D) drawn from the six-dataset reconstruction results of Section IV-A. Figure 9 then plots $M_z(t)$ and

Fig. 2. *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint's noiseless-simulator replay (center) and ground truth (left). The hardware image is visibly noisier — patch-boundary banding from per-patch readout variation is the dominant artifact — but the subject remains clearly recognizable.

TABLE V
HVK2D RESTRICTED PAIR-CORRELATION DIAGNOSTIC ACROSS FIVE SEEDS. DELIBERATELY FAVORABLE TO PAIR-OBSERVABLE STRUCTURE; EVIDENCE FOR ENTANGLEMENT-NECESSITY UNDER A MATCHED LINEAR READOUT.

Model	Mean MSE	Mean PSNR (dB)	Mean R^2
HVK2D entangling observables	8.5969×10^{-4}	32.77	0.9735
Freeze quantum	4.7904×10^{-3}	27.34	0.8533
No entanglement	3.1461×10^{-2}	15.03	0.0191
Raw-linear classical	3.1654×10^{-2}	15.00	0.0133
Parameter-matched classical	3.3033×10^{-2}	14.82	-0.0297
Random VQC	3.7616×10^{-2}	14.25	-0.1732
Freeze classical	6.0973×10^{-2}	12.17	-0.9025

$\mathcal{X}(t)$ for all four variants on the restricted diagnostic, with panel (c) zooming into the epoch-0–60 transition window: the entangling channel's susceptibility peak stands out clearly above every resource-matched classical control's, and its order parameter climbs to a substantially higher plateau. We then extend the same diagnostic to a representative sample of real images from all six datasets used in Section IV-A: **every one of 13 sampled images across all six datasets shows a detected phase-transition-like reorganization**, indicating this training-dynamics signature is a general property of HVK training on real image data, not an artifact of the synthetic restricted diagnostic. Full per-variant numbers, the multi-dataset breakdown, and per-dataset order-parameter trajectories are reported in the companion supplementary document.

V. Discussion

A. Task-Dependent Representational Scope

Whether a quantum feature map or kernel offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class, and dequantization results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched by classical algorithms once the right classical feature is identified [28]. The restricted diagnostic of Section IV-E is constructed so that the relevant structure is a genuine distant-element product raw or local features cannot represent under a fixed linear readout, and that is precisely the regime in which HVK's entangling channel earns its advantage. Natural image patches, at the resolution and patch size used in ordinary reconstruction, are a different regime — low-entanglement, locally structured data where a linear or shallow classical readout is already close to sufficient. The companion supplementary study's held-out ablation (Section V-B) characterizes exactly this second regime; the two results together delineate the boundary within which HVK's quantum components are and are not load-bearing.

B. Relationship to the Component-Wise Ablation Study

The results above establish what HVK is and several of its provable or verified properties. A separate, equally important question — under a resource-matched, freeze-based, multi-seed protocol, which components are strictly necessary for reconstruction quality on real, *held-out* images the model was not trained on — is answered in full in the companion supplementary document, *HVK Supplementary Study: Component-Wise Ablation, Leakage Audit, and Hardware Validation Methodology*. Its central finding: the trainable classical decoder is necessary and sufficient for held-out reconstruction quality, and no individual quantum component (trained circuit, entangling gates, Hamiltonian energy term) is isolated as necessary for that specific task; on held-out CIFAR-10, local-observable and raw-linear controls reach 18.80 ± 1.42 dB against HVK2D's 18.12 ± 1.54 dB (Wilcoxon $p = 9.5 \times 10^{-6}$, $n = 20$), with the same direction on five further real datasets. We report this as a rigorous complement, not a contradiction: it precisely delineates the boundary within which the

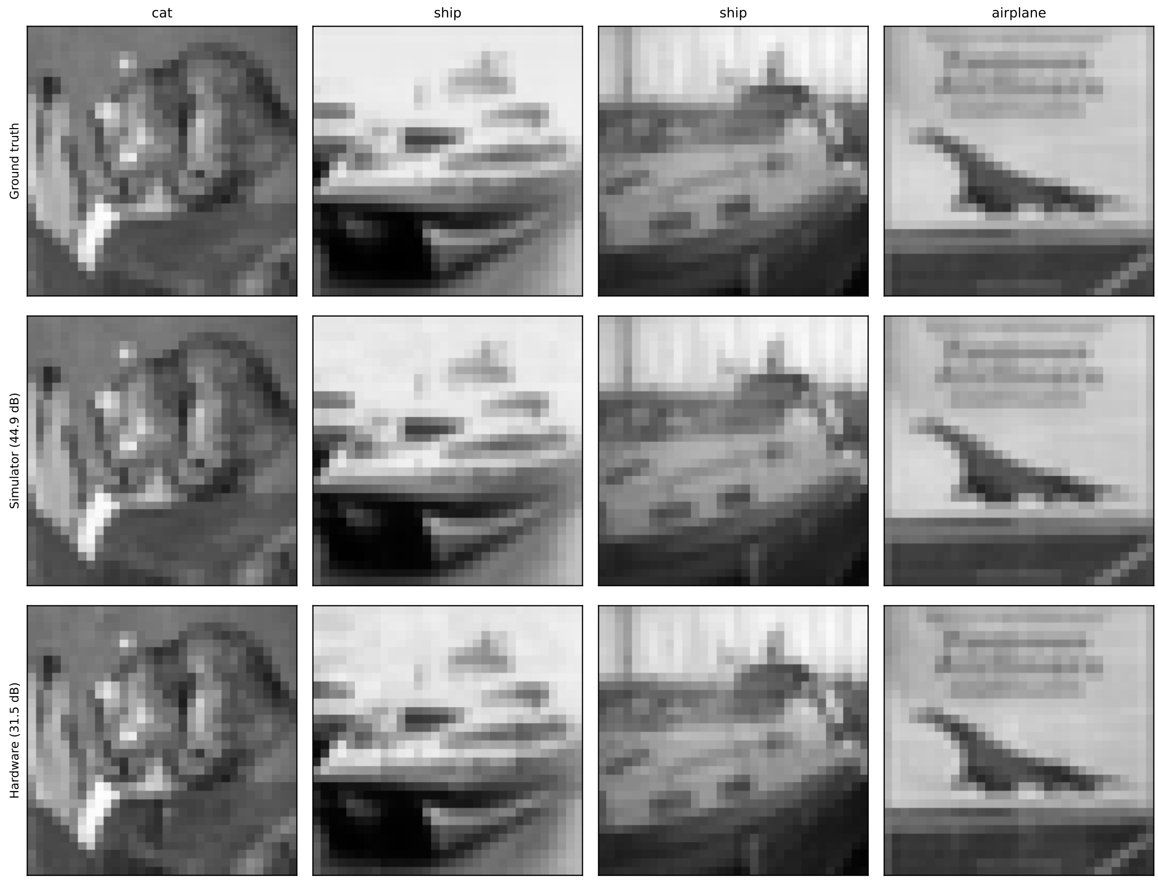

Fig. 3. Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained HVK2D checkpoints (middle row) and ground truth (top row).

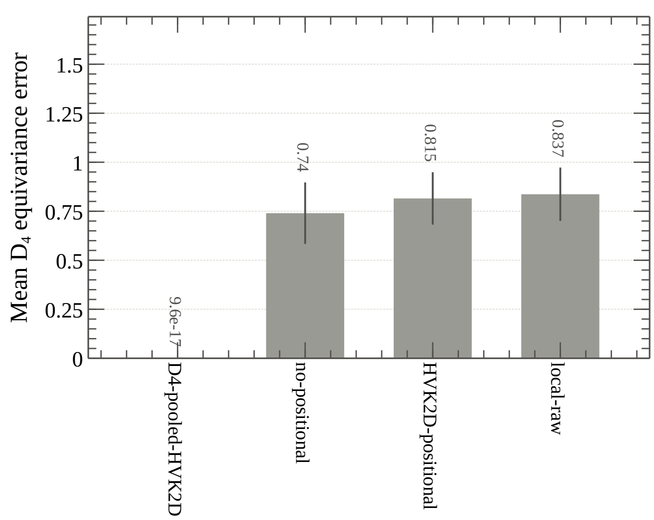

Fig. 4. D_4 equivariance diagnostic. The unpooled positional map is not equivariant; the group-pooled map is equivariant to numerical precision.

entanglement-necessity and provable-symmetry results above hold, and it caught a genuine circular-benchmark mistake of our own along the way, which the supplementary document documents as a transferable methodological lesson for this literature.

C. Topology Alignment as an Inductive Bias

Aligning the latent interaction graph with image-patch adjacency (HVK2D vs. HVK1D) is a plausible architectural inductive bias: on *Monalisa*, 2D HVK exceeds its 1D counterpart by 5.98 dB (40.70 vs. 34.72 dB). This is a single-image, single-seed observation, not a general principle, since the comparison is documented on one test case; we treat topology alignment as an inductive-bias hypothesis whose contribution has not yet been separated from decoder capacity, optimization budget, and dataset-specific structure.

D. Architectural Differentiation Matrix

Table VI contrasts HVK's structural properties against reference frameworks. Its distinguishing features — a Pauli-correlator latent space, 2D Fourier positional bias with enforced D_4 pooling, and a Hamiltonian energy diagnostic — are exactly the components whose provable/verified properties are established in Sections IV-D–IV-F.

E. Validated Scope and Next Steps

- Every reconstruction result in this paper (Sections IV-A–IV-B) is per-image trained, not held-out generalization;

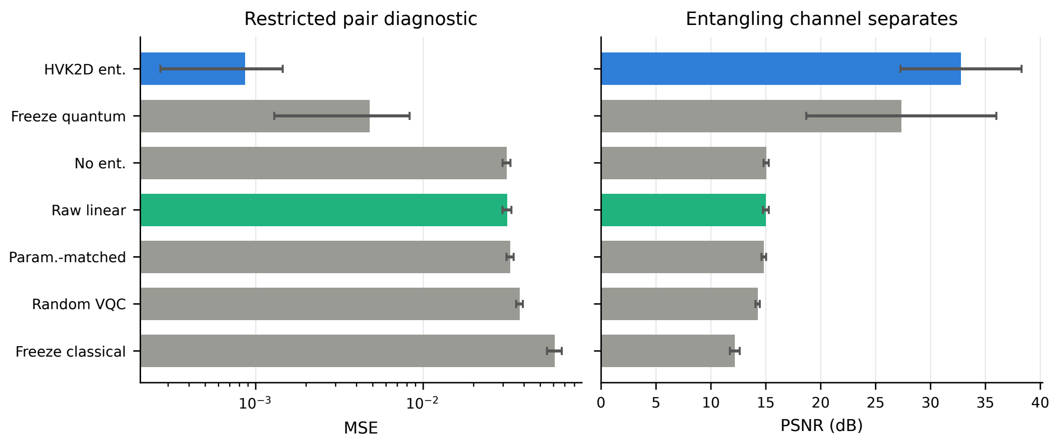

Fig. 5. Restricted pair-correlation diagnostic summary (mean $\pm$ std over five seeds). The entangling observable feature map dominates every control on this deliberately favorable task.

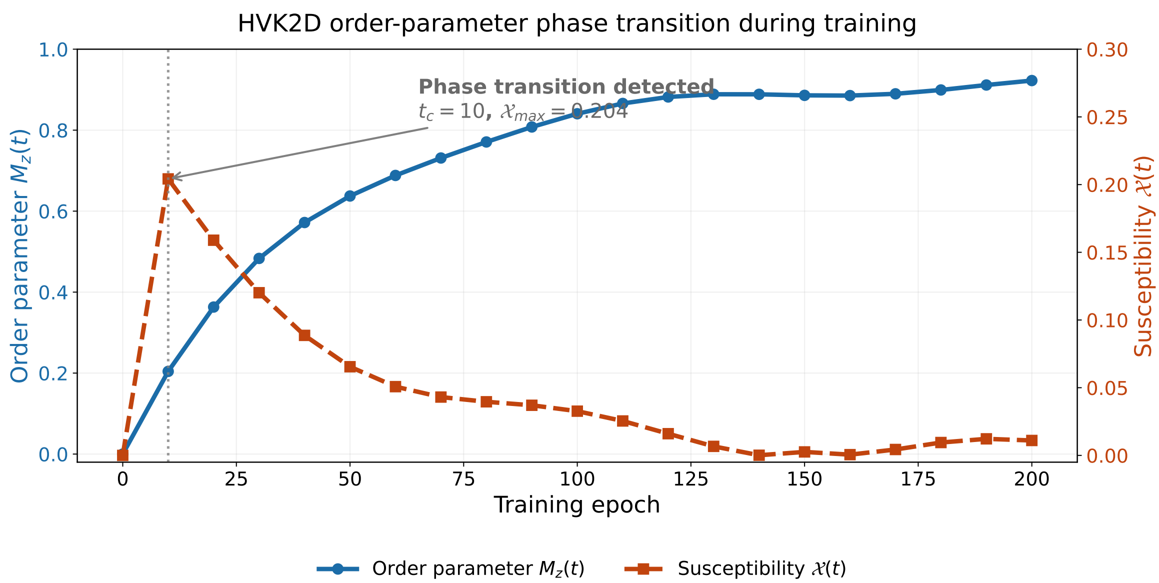

Fig. 6. Order-parameter phase transition for the HVK2D entangling baseline on the restricted pair-correlation diagnostic, plotted on a single shared training-epoch axis (x-axis) so the reorganization is visible without cross-referencing separate panels. The solid blue curve (left y-axis) is the order parameter $M_z(t)$, which rises from 0 toward a plateau near 0.9 as training proceeds. The dashed red curve (right y-axis) is the susceptibility $\mathcal{X}(t) = |M_z(t) - M_z(t - \Delta t)|$, the epoch-to-epoch rate of change of $M_z(t)$; it spikes sharply within the first few epochs and decays smoothly afterward. The vertical dotted gray line and annotated arrow mark the detected critical epoch $t_c = 10$, where the susceptibility reaches its maximum value $\mathcal{X}_{\max} = 0.204$ — the point of most abrupt reorganization in the learned observable field, per the definitions of $M_z(t)$ and $\mathcal{X}(t)$ in Eqs. (2)–(3).

the companion supplementary study's held-out protocol is the appropriate evidence for a generalization claim, and its central finding (classical decoder necessary and sufficient) should be read alongside the architecture-level results here.

- The D_4-pooled map (Section IV-D) is currently a feature-grid-level construction and numerical diagnostic, not yet integrated into the trainable end-to-end reconstruction pipeline; doing so is concrete future work (Section VI).
- The entanglement-necessity result (Section IV-E) is scoped to a task deliberately constructed to require nonlocal structure; it does not by itself establish an advantage on ordinary natural-image reconstruction (see Section V-A).
- The hardware pilot (Section IV-C) is a five-image, single-trial, unmitigated pilot, not a statistically resolved hardware benchmark.
- The phase-transition multi-dataset extension (Section IV-F) samples 2–3 images per dataset, single-seed; it is a directional diagnostic at that scope, not a statistically resolved claim.
- Topology alignment (Section V-C) is documented on a single test case, not a multi-seed comparison.

VI. Conclusion

We presented the Hamiltonian Vision Kernel, a physics-informed hybrid quantum–classical image autoencoder distinguished by a Pauli-correlator latent space, an enforced and

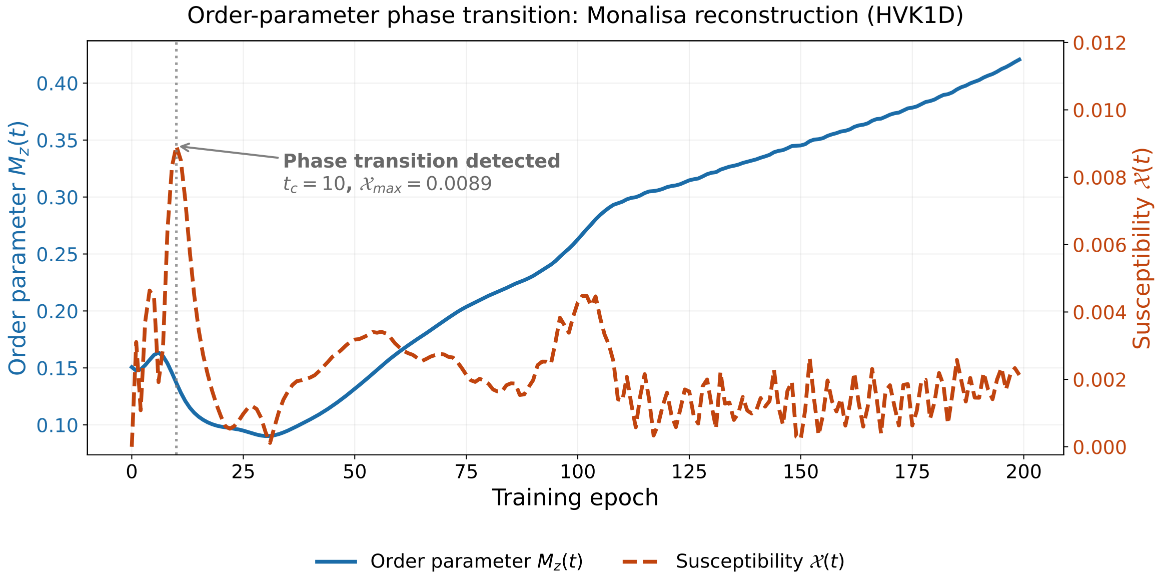

Fig. 7. Order-parameter phase transition for the actual *Monalisa* reconstruction run (HVK1D, the same shared-baseline checkpoint used throughout this paper), not the synthetic restricted diagnostic. Curves and axes follow the same convention as Figure 6: order parameter $M_z(t)$ in solid blue (left axis), susceptibility $\mathcal{X}(t)$ in dashed red (right axis). The detected critical epoch is $t_c = 10$ with peak susceptibility $\mathcal{X}_{\max} = 0.0089$, an order of magnitude smaller in absolute susceptibility than the synthetic diagnostic of Figure 6 but still a statistically sharp reorganization by the same median-plus-two-standard-deviation threshold; the order parameter itself continues to climb well past t_c as training proceeds toward the final 40.70 dB reconstruction reported in Section IV-A.

TABLE VI
ARCHITECTURAL DIFFERENTIATION MATRIX

Axis	Classical AE	GAN AE	Standard QAE	HVK (Ours)
Latent Space	Continuous $\mathbb{R}^d$	Continuous $\mathbb{R}^d$	Quantum Register	Pauli Correlator Vector
Spatial Bias	Implicit Convolutions	Implicit Convolutions	None	2D Fourier + enforced D_4 pooling
Regularizer	Bottleneck Size	Discriminator Loss	Swap-Test Fidelity	Hamiltonian energy diagnostic
Group Alignment	None	None	Reference Swaps	SO(3) Cartan motivation + enforced D_4 pooling
Target Task	Reconstruction	Reconstruction	Compression	Spatially Aware Reconstruction

numerically verified D_4-equivariant observable-pooling extension, and a learnable Heisenberg energy diagnostic. Independently of reconstruction quality, we established three verified structural properties: a provable D_4 symmetry construction (equivariance error 9.57×10^{-17}), an entanglement-necessity result on a task requiring nonlocal structure ($R^2 = 0.9735$ vs. $R^2 \leq 0.02$ for non-entangling controls), and a training-dynamics phase-transition signature detected in every one of 13 sampled images across six real datasets. We further reported a real IBM Quantum hardware reconstruction pilot — an already-trained checkpoint's exact parameters replayed with no retraining reach 25.90–31.52 dB PSNR directly from hardware measurements, at negligible quantum-processor cost — a first, honest hardware validation for this architecture family.

A companion supplementary study reports a separate, rigorous question in full: under a resource-matched, freeze-based ablation protocol, which components are necessary for reconstruction quality on held-out real images. Its finding — the classical decoder is necessary and sufficient, and no individual quantum component is isolated as necessary for that task — is not in tension with the results here; it delineates precisely where entanglement and symmetry structure matter (tasks requiring nonlocal structure) and where, on ordinary natural-image reconstruction, a well-matched classical readout is already close to sufficient. We consider this boundary, established rigorously rather than asserted, to be as much a contribution of this research program as the architecture itself.

Future work should integrate the D_4-pooled map into an end-to-end trainable pipeline (currently a feature-grid-level construction, Section V-E); scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials; search for further real-world tasks with genuine nonlocal structure where HVK's entanglement-necessity property translates into a reconstruction advantage; and extend the resource-matched, leakage-audited evaluation protocol used in the companion supplementary study to other hybrid quantum vision architectures.

ACKNOWLEDGEMENTS

We thank the maintainers of the open-source scientific computing libraries used in this work.

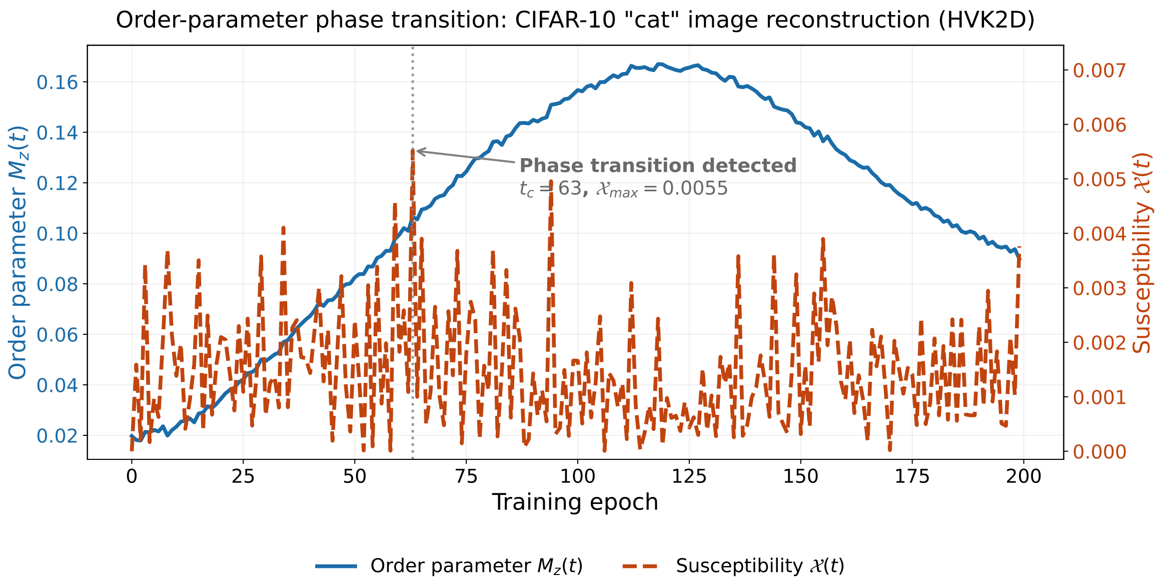

Fig. 8. Order-parameter phase transition for a specific, named CIFAR-10 image — the *cat* image (HVK2D) from the same-set reconstruction results of Section IV-A — again using the actual reconstruction-training run rather than the synthetic restricted diagnostic. Curves and axes follow the same convention as Figure 6. The detected critical epoch is $t_c = 63$ with peak susceptibility $\mathcal{X}_{\max} = 0.0055$; the order parameter rises through the transition and continues climbing to a peak around epoch 115 before gradually declining, illustrating that the training-dynamics reorganization on a real natural image need not occur early in training or monotonically approach a plateau, unlike the synthetic diagnostic or the *Monalisa* run of Figure 7.

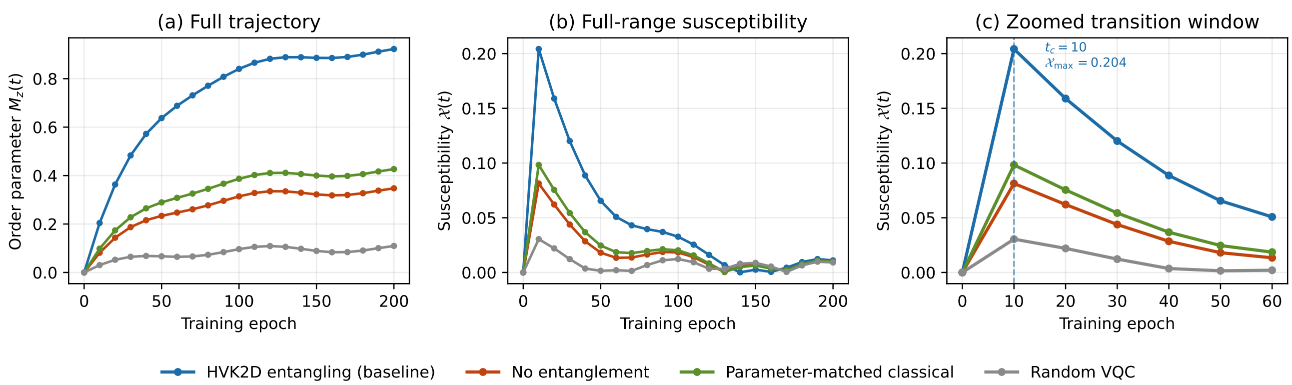

Fig. 9. Order-parameter phase-transition diagnostic on the restricted pair-correlation task. (a) Full order-parameter trajectory $M_z(t)$ and (b) full-range susceptibility $\mathcal{X}(t)$; (c) zooms into the epoch-0–60 transition window, where the entangling channel's susceptibility peak separates cleanly from every classical control.

Appendix

For a one-dimensional chain of n_{bonds} bonds indexed by i and sites $(i, i+1)$, the anisotropic Heisenberg Hamiltonian is

$$H_{\text{Heis}} = \sum_{i=0}^{n_{\text{bonds}}-1} \Big(J_x^{(i)} X_i X_{i+1} + J_y^{(i)} Y_i Y_{i+1} + J_z^{(i)} Z_i Z_{i+1} \Big), \tag{4}$$

where X, Y, Z are Pauli operators and $J_\alpha^{(i)}$ bond-resolved couplings. In the isotropic limit ($J_x = J_y = J_z$) the model has exact SU(2) symmetry; for $J_x = J_y \neq J_z$ (XXZ) the U(1) symmetry generated by $\sum Z_i$ is preserved; for generic couplings (XYZ) this breaks to $\mathbb{Z}_2 \times \mathbb{Z}_2$ [15]. HVK fixes six sites and treats every $J_\alpha^{(i)}$ as a learnable parameter, $J_\alpha^{(i)} \sim \mathcal{N}(0, 0.01)$. Given patch n's measured observables, the Heisenberg energy is

$$\mathcal{E}_n(\theta, J) = \sum_{i=0}^{4} \sum_{\alpha \in \{x,y,z\}} J_\alpha^{(i)} \langle \sigma_\alpha^{(i)} \sigma_\alpha^{(i+1)} \rangle_n, \tag{5}$$

and the patch-averaged training energy loss is $\mathcal{L}_{\text{energy}} = \frac{1}{N} \sum_n \mathcal{E}_n$. The total objective combines pixel fidelity with this regularizer,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda\, \mathcal{L}_{\text{energy}}, \quad \mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{n=1}^{N} \big\| D(o_n, e_n) - P_n \big\|_F^2, \tag{6}$$

with D the classical decoder and P_n the ground-truth patch.

Per-patch coordinates (r_i, c_j) are mapped to an 8-D Fourier positional vector, $e_{4k} = \sin(\omega_k \pi r_i)$, $e_{4k+1} = \cos(\omega_k \pi r_i)$, $e_{4k+2} = \sin(\omega_k \pi c_j)$, $e_{4k+3} = \cos(\omega_k \pi c_j)$ for $k \in \{0, 1\}$ [16], projected via a learned $W_{\text{pos}} \in \mathbb{R}^{6\times 8}$ to $\theta_{\text{position}} =$

$W_{\text{pos}}e \in \mathbb{R}^6$. The 6-qubit circuit applies, in order: angle embedding $U_{\text{enc}}(x_n) = \bigotimes_{i=0}^{5} R_X(x_n^{(i)})$; positional modulation $U_{\text{pos}} = \bigotimes_{i=0}^{5} R_Y(\theta_{\text{position}}^{(i)})$; and $L = 2$ strongly entangling layers $U_{\text{var}}(\theta) = \prod_{l=1}^{L}\left[\text{CNOT-Ring} \cdot \bigotimes_{i=0}^{5} \text{Rot}(\phi_i^l, \theta_i^l, \omega_i^l)\right]$, giving $|\Psi_n\rangle = U_{\text{var}}U_{\text{pos}}U_{\text{enc}}|0\rangle^{\otimes 6}$.

The Heisenberg correlators define three observable sectors $\Xi_{ZZ}, \Xi_{XX}, \Xi_{YY}$ embedded in $\mathbb{R}^{27}$. Two symmetry reductions act sequentially: the per-qubit positional rotations belong to $\mathrm{SU}(2)$ (a double cover of spatial $\mathrm{SO}(3)$ Bloch rotations) and break qubit-site permutation symmetry down to the Cartan sector; and the 4×4 patch grid carries the dihedral group $D_4 = \langle r, s \mid r^4 = s^2 = (sr)^2 = e\rangle$, under which observable vectors transform as $o_{g(i,j)} \approx \Phi_g\, o_{i,j}$, decomposing the latent space as $\mathcal{V}_{\text{obs}} \approx m_A A_1 \oplus m_B A_2 \oplus m_C B_1 \oplus m_D B_2 \oplus m_E E$. This motivates the reduction $\mathrm{SU}(2)$-isotropic $\xrightarrow{\text{positional encoding}}$ Cartan $\xrightarrow{\text{grid geometry}}$ D_4, which the unpooled map only motivates rather than enforces (Section IV-D gives the pooling construction that does enforce it).

Adam optimizer, $\eta = 0.003$ (1D) / $\eta = 0.004$ (2D); energy regularization $\lambda = 0.01$ unless explicitly ablated; initialization scales follow established barren-plateau-safe limits [1], [22]. The MPS feature extraction unrolls each 64×64 patch to a 4096-dimensional vector (2^{12}), reshaped to a 12-site spin state and compressed by sequential SVD to bond dimension $\chi = 4$, yielding 24 local Pauli expectations, 11 nearest-neighbor ZZ correlations, and 11 bipartite entanglement entropies ($S_i = -\sum_k p_k^{(i)} \log(p_k^{(i)} + \varepsilon)$, $\varepsilon = 10^{-8}$) — a 46-D feature vector, linearly compressed to 6-D before circuit injection.

The entanglement-necessity diagnostic of Section IV-E was independently audited for target–feature leakage: the target is a fixed synthetic function of simulated patch features not concatenated into any candidate model's own input block, so no model can achieve fit quality by copying precomputed target-construction quantities from its own features. This audit was itself the product of catching an earlier, circular benchmark of our own (an entangling observable channel reaching a spurious $R^2 \approx 1$ by copying, not learning, distant-patch product terms that had been concatenated onto its own input); the full post-mortem and the resulting general-purpose leakage-detection rule are given in the companion supplementary document.

Table VII and Figure 10 report circuit resources for the HVK1D and HVK2D patch circuits compiled to an IBM Heron-class basis (patch index 0). Both variants use six measured qubits and depth-18 circuits after transpilation; HVK2D requires more CNOT gates because it encodes additional grid-adjacency links. This is a compiled-circuit feasibility check across the broader variant sweep, complementary to the finite-shot reconstruction pilot of Section IV-C: this appendix draws no image-quality conclusion by itself, while Section IV-C reports the actual hardware-measured reconstruction numbers, with full methodology in the companion supplementary document.

TABLE VII
CIRCUIT-RESOURCE SUMMARY FOR HVK PATCH CIRCUITS MAPPED TO AN IBM HERON-CLASS BASIS.

Variant	Qubits	Depth	R_Y	R_Z	CNOT
HVK1D	6	18	11	6	10
HVK2D	6	18	13	6	14

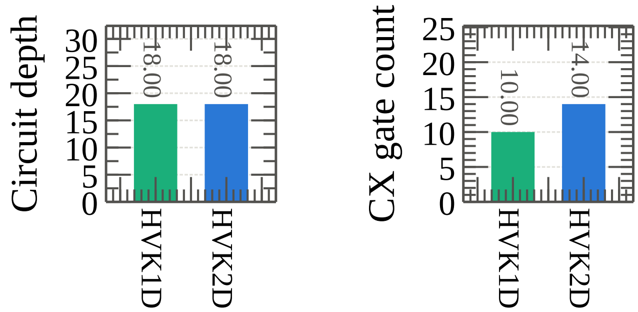

Fig. 10. IBM Cloud circuit-resource summary for the HVK1D and HVK2D hardware probes.

REFERENCES

[1] M. Cerezo et al., "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, no. 9, pp. 625–644, 2021.

[2] J. Tilly et al., "The Variational Quantum Eigensolver: A review of methods and best practices," *Physics Reports*, vol. 986, pp. 1–128, 2022.

[3] B. Bauer, S. Bravyi, M. Motta, and G. K.-L. Chan, "Quantum Algorithms for Quantum Chemistry and Quantum Materials Science," *Chemical Reviews*, vol. 120, no. 22, pp. 12685–12717, 2020.

[4] S. Chakraborty, S. B. Mandal, and S. H. Shaikh, "Quantum image processing: challenges and future research issues," *International Journal of Information Technology*, vol. 14, no. 1, pp. 475–489, 2018.

[5] R. Zhou, W. Hu, P. Fan, and H. Ian, "Quantum realization of the bilinear interpolation method for NEQR," *Scientific Reports*, vol. 7, no. 1, p. 2511, 2017.

[6] X. Yao et al., "Quantum Image Processing and Its Application to Edge Detection: Theory and Experiment," *Physical Review X*, vol. 7, no. 3, p. 031041, 2017.

[7] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete, "Matrix product states and projected entangled pair states: Concepts, symmetries, theorems," *Reviews of Modern Physics*, vol. 93, no. 4, p. 045003, 2021.

[8] J. I. Latorre, "Image compression and entanglement," arXiv:quant-ph/0510031, 2005.

[9] Z. Han, J. Wang, H. Fan, L. Wang, and P. Zhang, "Unsupervised Generative Modeling Using Matrix Product States," *Physical Review X*, vol. 8, no. 3, p. 031012, 2018.

[10] S. M. Fei, "Compressed Sensing Based on Tensor Network Machine Learning," *Physical Science & Biophysics Journal*, vol. 5, no. 1, p. 16000174, 2021.

[11] D. Guala et al., "Practical overview of image classification with tensor-network quantum circuits," *Scientific Reports*, vol. 13, no. 1, p. 4427, 2023.

[12] A. Haga, "Quantum annealing-based computed tomography using variational approach for a real-number image reconstruction," *Physics in Medicine and Biology*, vol. 69, no. 4, p. 02155, 2024.

[13] X. Zhai, T. Xiao, J. Huang, J. Fan, and G. Zeng, "Quantum neural compressive sensing for ghost imaging," *Physical Review Applied*, vol. 23, no. 1, p. 014018, 2025.

[14] M. A. Nau, A. H. Vija, W. Gohn, M. P. Reymann, and A. Maier, "Exploring the Limitations of Hybrid Adiabatic Quantum Computing for Emission Tomography Reconstruction," *Journal of Imaging*, vol. 9, no. 10, p. 221, 2023.

[15] B. Craps, M. D. Clerck, O. Evnin, and P. Hacker, "Integrability and complexity in quantum spin chains," *SciPost Physics*, vol. 16, no. 2, p. 041, 2024.

[16] R. Xu, X. Wang, K. Chen, B. Zhou, and C. C. Loy, "Positional Encoding as Spatial Inductive Bias in GANs," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 13564–13573.

[17] J. Y. Araz and M. Spannowsky, "Classical versus quantum: Comparing

tensor-network-based quantum circuits on Large Hadron Collider data," *Physical Review A*, vol. 106, no. 6, p. 062423, 2022.

[18] D. Slabbert and F. Petruccione, "Hybrid Quantum-Classical Feature Extraction approach for Image Classification using Autoencoders and Quantum SVMs," arXiv:2410.18814, 2024.

[19] Z. Liu, P.-X. Shen, W. Li, L.-M. Duan, and D.-L. Deng, "Quantum capsule networks," *Quantum Science and Technology*, vol. 8, no. 1, p. 015016, 2022.

[20] J. Shi, W. Wang, X. Lou, S. Zhang, and X. Li, "Parameterized Hamiltonian Learning With Quantum Circuit," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 5, pp. 6086–6095, 2022.

[21] J. Preskill, "Quantum Computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[22] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Communications*, vol. 9, no. 1, p. 4812, 2018.

[23] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.

[24] E. M. Stoudenmire and D. J. Schwab, "Supervised Learning with Tensor Networks," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.

[25] T. Cohen and M. Welling, "Group Equivariant Convolutional Networks," in *Proceedings of the 33rd International Conference on Machine Learning*, PMLR, vol. 48, 2016, pp. 2990–2999.

[26] J. J. Meyer, M. Mularski, E. Gil-Fuster, A. A. Mele, F. Arzani, A. Wilms, and J. Eisert, "Exploiting Symmetry in Variational Quantum Machine Learning," *PRX Quantum*, vol. 4, p. 010328, 2023.

[27] H.-Y. Huang, R. Kueng, and J. Preskill, "Predicting many properties of a quantum system from very few measurements," *Nature Physics*, vol. 16, pp. 1050–1057, 2020.

[28] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, and J. R. McClean, "Power of data in quantum machine learning," *Nature Communications*, vol. 12, no. 1, p. 2631, 2021.